

Entanglement-Induced Resilience in Analog Simulation and Quantum Control

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Abstract. We identify and numerically demonstrate entanglement-induced resilience in realistic quantum dynamics. Our main evidence comes from three application settings: analog simulation of spin models, analog simulation of fermionic models, and quantum-control protocols. Across extensive simulations of transverse-field Ising systems, Fermi-Hubbard lattices, and quantum-dot gate control, states that generate larger subsystem entanglement consistently show smaller coherent dynamical errors. Theoretically, local perturbative errors are bounded by a Frobenius-norm contribution plus entropy-dependent corrections, explaining why highly entangled dynamics approaches average-case rather than worst-case behavior. These results suggest a practical route to more robust analog simulators and control protocols.

Keywords: analog quantum simulation, fermionic simulation, quantum control, entanglement

1 Introduction

Analog quantum simulators and quantum-control devices are built from interacting many-body systems, and their dominant errors are often coherent: static disorder, imperfect couplings, crosstalk, pulse distortions, and spectator-induced shifts. Such errors are hard to remove completely, especially in near-term devices where full quantum error correction is unavailable. Our work asks whether the many-body dynamics itself can reduce sensitivity to these perturbations.

The central answer is that dynamically generated entanglement can provide a native resilience mechanism. The main contribution of this work is not only a general bound, but a set of numerical demonstrations in three experimentally motivated settings: spin-model analog simulation, fermionic analog simulation, and quantum control. In each setting, we compare high-entanglement and low-entanglement dynamical regimes and observe the same behavior: larger subsystem entanglement suppresses coherent dynamical error.

2 Theory

Let $H_0(t)$ generate the ideal evolution $U_0(t)$ and let the implemented dynamics be generated by

$$\tilde{H}(t) = H_0(t) + H_{\text{pert}}(t),$$

where H_{pert} contains disorder and systematic imperfections. For an initial state $|\psi\rangle$, the state-dependent error obeys

$$\|(U_0(t) - U(t))|\psi\rangle\| \leq \int_0^t d\tau \|H_{\text{pert}}(\tau)|\psi(\tau)\rangle\|, \quad (1)$$

where $|\psi(\tau)\rangle = U_0(\tau)|\psi\rangle$. Therefore, the relevant error is not only the operator norm of the perturbation, but also its action on the actual evolved state.

For local perturbations $H_{\text{pert}} = \sum_j H_j$, this action is controlled by local entanglement. If $A = \sum_j A_j$ is positive semidefinite and local, then

$$\langle\psi|A|\psi\rangle \leq \frac{\text{Tr}(A)}{d} + \sum_j \|A_j\| \sqrt{2 \log d_{\text{supp}(A_j)} - 2S(\rho_j)}, \quad (2)$$

where ρ_j is the reduced state on the support of A_j . Applying this to $A = H_{\text{pert}}^\dagger H_{\text{pert}}$ shows that highly entangled subsystems drive the entropy-dependent correction toward zero, leaving the normalized Frobenius scale $\|H_{\text{pert}}\|_F$. Thus entanglement changes the relevant scaling from worst-case spectral-norm behavior to average-case Frobenius-norm behavior.

3 Application I: Spin Models

Our first and most direct numerical test is analog simulation of spin systems. We study the one-dimensional transverse-field Ising model

$$H_0 = h_x \sum_i X_i + h_y \sum_i Y_i + J \sum_i X_i X_{i+1},$$

with $h_x = 0.809$, $h_y = 0.9045$, and $J = 1$. The imperfect simulator includes both local disorder, $H_{\text{dis}} = \sum_i \delta_i X_i$, and systematic coupling imperfection, $H_{\text{imp}} = \eta \sum_i X_i X_{i+1}$.

The numerical results show a clear separation between typical and atypical input states. For $|0\rangle^{\otimes N}$, the dynamics rapidly generates subsystem entanglement; after this growth, the simulation error follows the Frobenius-norm prediction. For $|+\rangle^{\otimes N}$, entanglement remains small and

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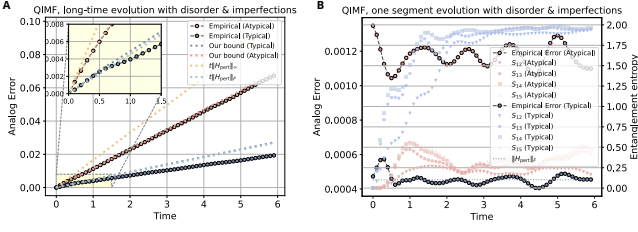


Figure 1: **Spin-model analog simulation.** In the transverse-field Ising model, high-entanglement dynamics follows the Frobenius-norm estimate, while low-entanglement dynamics remains closer to the worst-case spectral-norm behavior.

the error stays much closer to the spectral-norm bound. We also evaluate one-segment simulation errors along the trajectory and find that local entropy growth is accompanied by error suppression. Additional spin simulations in two-dimensional settings show the same trend.

4 Application II: Fermionic Models

The second application is analog simulation of fermionic systems, where the many-body Hilbert space and conservation laws differ substantially from spin models. We study Fermi-Hubbard dynamics with periodic boundary conditions,

$$H_0 = - \sum_{i,\sigma} (a_{i,\sigma}^\dagger a_{i+1,\sigma} + \text{H.c.}) + V \sum_i n_{i,\uparrow} n_{i,\downarrow},$$

and perturb the simulator by an additional Coulomb term

$$H_{\text{pert}} = \delta \sum_i n_{i,\uparrow} n_{i,\downarrow}.$$

We compute segment-level errors with $\Delta t = 0.1$ for several initial fermionic configurations. The results again show a robust inverse correlation: when two-site subsystem entanglement approaches its maximum, the empirical simulation error drops toward the Frobenius estimate; when entanglement decreases, the error develops sharp spikes. This behavior appears in both one-dimensional and two-dimensional Fermi-Hubbard simulations, indicating that entanglement-induced resilience is not an artifact of Pauli spin dynamics.

5 Application III: Quantum Control

The third application is real-time quantum control. We consider a target gate implemented in a larger register, where nearby spectator qubits couple to the driven qubit and create unavoidable coherent perturbations. For a target subsystem A consisting of the driven qubit and nearby spectators, our bound predicts that entanglement between A and the rest of the register reduces the effective gate error toward the Frobenius scale of the time-dependent perturbation.

We test this prediction in a 3×4 quantum-dot array using robust control pulses for X_π , $X_{\pi/2}$, and $X_{2\pi}$ gates. The perturbation model includes frequency drift, residual

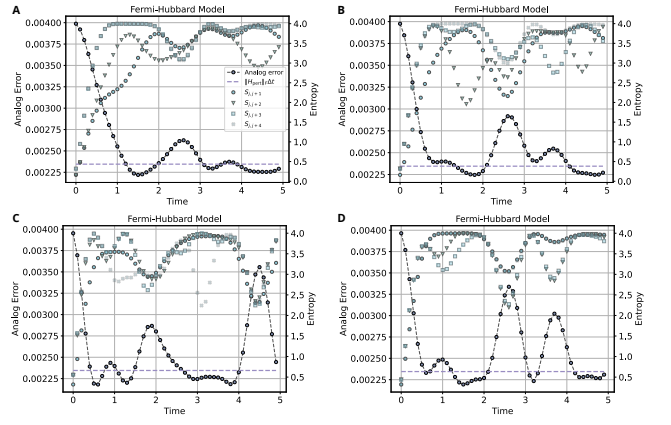


Figure 2: **Fermionic analog simulation.** In Fermi-Hubbard simulations, segment errors decrease as the relevant two-site entanglement entropy increases, and error spikes occur when the entropy drops.

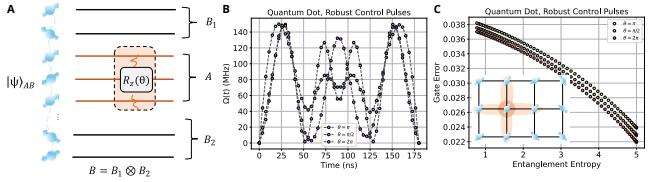


Figure 3: **Quantum-control protocols.** For robustly driven quantum-dot gates, larger entanglement between the target neighborhood and the rest of the register correlates with lower effective gate error.

couplings, and pulse-amplitude errors. To scan different entanglement regimes, we use purified Gibbs states whose temperature controls the entanglement entropy between the target neighborhood and the rest of the array. In all tested gates, the effective error decreases rapidly as this entropy approaches its maximum. This shows that entanglement can improve gate robustness even on top of optimized robust-control pulses.

6 Diagnostics and Outlook

The numerical studies also suggest a practical diagnostic. Even when the precise coefficients of H_{pert} are unknown, the local structure of dominant noise terms is often known from the device layout. Measuring coefficient-independent cross-term correlators,

$$\frac{\langle \psi(t) | H_j^\dagger H_{j'} + \text{H.c.} | \psi(t) \rangle}{\|H_j\| \|H_{j'}\|},$$

can indicate whether the relevant local reduced states are close to maximally mixed. In our spin-model tests, typical high-entanglement states yield many near-zero cross terms, while atypical low-entanglement states yield sizable correlations.

Taken together, the spin-model, fermionic-model, and quantum-control experiments support a unified picture: entanglement growth can make coherent perturbations act like average-case errors rather than worst-case errors.

This provides concrete guidance for designing analog simulators and control protocols: engineer dynamics that generate entanglement on the supports where dominant perturbations act.

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